
Monday 13th January 2025

To-Do Today:

- Research papers and find a sub-topic of interest within pulsar timings
- Think of a question that could be answered within 8 weeks of researching

Work Done Today:

I began looking into papers about various topics that were listed on the guide and through ads. I looked into topics such as parameters of binary pulsar systems, planetary systems around pulsars and gravitational waves released from binary pulsars.

I dove into the papers and was caught by the discovery of the first exo-planet which was found using pulsars. The papers ([Wolszczan and Frail 1992](#), [Konacki and Wolszczan 2003](#)) described their methods and results about the planets orbiting a pulsar PSR1257 + 12.

In these papers they demonstrated that they found the two exoplanets through various plots such as the average periodicity of the pulse, and they took into account the slight delay on the time of arrival with each reading when taking the mean.

I wanted to try and replicate an experiment that was done already, from research online I found this website which gave a breakdown of exoplanets found from pulsar by [Macy Huston](#) and found that only 8 exoplanets have been found by using pulsar timings over the last 30 years, as shown in the [NASA exoplanet archive](#).

To make sure that we finish on time within the 8 weeks we decided that we would choose a system that only contains 1 exoplanet and 1 pulsar as we want to simplify our calculations and find more about the planet. This means that our sample size halves as 3 planets found using pulsar timing belong to the same system (PSR 1257 + 12) and another one was found in a binary star system between a pulsar and dwarf star.

For Next Time:

From today's session we managed to look into subtopics for pulsars and found papers relating to exo-planet discoveries using the pulsar timing method. For next time we decided that we need to:

- Finalise on a topic after researching a bit more and be certain that trying to rediscover an exo-planet is a feasible topic for us to investigate.
- Once we finalise the topic we want to try and determine a question that we could answer.

Thursday 16th January 2025

To - do Today:

- Finalise on a topic after researching a bit more papers and be certain that trying to rediscover an exo-planet is a feasible topic for us to investigate.
- Once we finalise the topic we want to try and determine a question that we could answer.

Work Done Today:

Today, I carried on researching more papers about how exoplanets are found using pulsar timings. I used the NASA exoplanet archive previously mentioned to look into how each one was found trying to avoid systems in binary stars and multiple planets. The only plausible systems were PSR B0329+54 and PSR J1719-1438 which seemed to meet these conditions.

I read through the paper written by [Starovoi & Rodin 2017](#) which described their methods. However, this wasn't very clear to me but I managed to understand the general idea. So, to clarify how pulsar timing is used to find exoplanets in general, I watched a video which helped deepen my understanding.

The [youtube video](#), made by Ben, explained the concepts of how a system of a pulsar and planet work. They essentially orbit a common centre of mass, and depending on the planet's position the light emitted is delayed or arrives quicker to us. We can use this delay and plot this, in theory it should give us a sine wave which would tell us about the period of the planet. This idea seemed simple enough for us to compute, we would need a data set about a pulsar with enough timings to plot 1 period of a planet.

So to move forward, we had to find sources of data we could use for the coming weeks. This could be data such as flux, residuals etc. We just had to find something we could use to investigate and find a planet.

Looking through papers and other online research it was proving difficult for us to find a suitable set of data that we could use for any pulsar. Having to look through papers to see which telescope they used and whether they linked any data we could use was close to 0. We found one set of data for PSR J1719-1438 from a paper by [Balies](#) where they linked some data they used in order to confirm the existence of an exotic planet orbiting a pulsar.

For Next Time:

In total, today we successfully managed to finalise our topic to exoplanet discovery using the pulsar timing method. We briefly discussed the kind of question we could answer, this was along the lines of "how reproducible is finding a planet using pulsar timing" or "can we find this planet using pulsar timing".

For next time we decided that we need to:

- Finalise the question that we're going to answer, so we have a clear aim to work towards but also not limited to experiment more.
- Start gathering ideas for the preliminary report such as papers used, data that we found and important equations.
- Get familiar with the data set that we have and see how the files are used in scientific discovery.

Monday 20th January 2025

To - do Today:

- Finalise the question that we're going to answer, so we have a clear aim to work towards but also not limited to experiment more by having a discussion with the demonstrator

- Start gathering ideas for the preliminary report such as papers used, data that we found and important equations.
- Mess around with the data and try and see if we can manage extracting useful information from it and create some toy plots.

Work Done Today:

We started off by discussing our plans for today, we realised that we spent a lot of time researching and scouring the internet that we needed to try and work a little bit more on the project. We first started by finalising the question by talking to our demonstrator about how feasible the idea was. Ultimately we decided, with help, that the most feasible idea for us to work on would be something along the lines of "Can we rediscover a certain exoplanet and its properties which is orbiting PSR####". This doesn't limit our ability to explore the pulsar first and then explore the planet. The main goal is to find that the pulsar has a planet and find its period. This question doesn't limit us to find out more about the planet if we have more time such as the mass, size, etc.

Moving forward we wanted to experiment with the data that we found last session from a paper about PSR J1719-1438 ([Bailes, Matthew, et al](#)). We wanted to explore packages in python and other code that could help us. The data given to us included the parameters of the pulsar (.par file) and the arrival times(.tim file). This data is supposed to help derive some further properties about the pulsar such as the age, distance and then later find the exoplanet.

We spent the rest of the session researching how the files are used to discover properties about the pulsar. We found that these files are manipulated using Tempo2. This analyses the data but uses fortran and C as its code which is pretty hard for us to learn. So instead we found libstempo which is a wrapper for Tempo2 and helps us code in python instead. We spend this session looking into this and trying to install it. It took longer than expected and after having asked the demonstrator we finally got it installed.

For Next Time:

Ultimately, we managed to finalise our topic and question, we managed to narrow our question down to roughly "Can we, using pulsar timing, discover the exotic exoplanet orbiting PSR J1719 - 1438?". We also began to investigate how the data we have can be used and what steps are needed for us to get plots and characteristics.

For next time we decided that we need to:

- Experiment with the data, since we ran out of time installing libstempo, next time we can try out tutorials on basic functions.
- Continue to gather data, figures, plots and papers to put together for the preliminary report

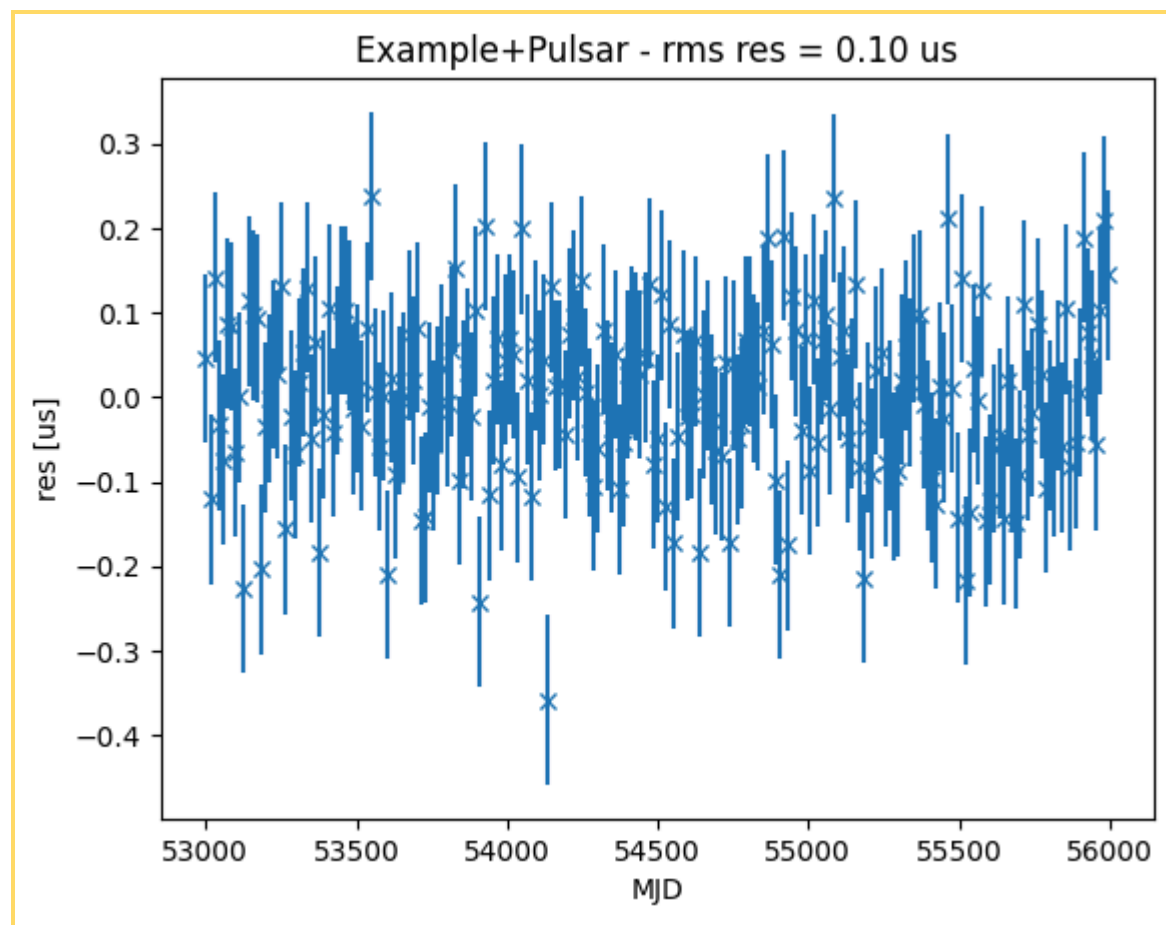
Thursday 23rd January 2025**To - do Today:**

- Carry on from monday's work and work with simulated data to make pulsars and see how the libstempo module works
- End up with some sort of plan to use the data we have and how the wrapper will help us determine characteristics about the pulsar and planet
- Start planning/writing out the preliminary report

Work Done Today:

Carrying on from last time, after having just installed condacolab and then enterprise pulsar which includes libstempo. As mentioned before, libstempo is a key python wrapper which helps work with the tempo2 pulsar package. The data that we sourced from Bailes, Matthew, et al was told to be processed into tempo2. Unfortunately, tempo2 is an old software running on fortran and C, which none of us know how to use, that's where libstempo comes in.

During the session I continued to experiment with the example tutorial of the package. This was found on [github](#) which was working on gravitational waves. From this I followed the steps which were messing around with creating my own .tim and .par files and simulating a fit for a pulsar. Eventually, creating plots of a fake pulsar with simulated errors on the parameters and finding out the energy lost and the magnetic field of the pulsar.



Some important equations I learned from this ([Appendix 3 & 5](#)), given by [Lorimer & Kramer 2008](#). These equations give the energy lost and the surface magnetic field respectively. Where P and $\dot{P}$ is the period and period derivative respectively of the pulsar, and I is the moment of inertia of the pulsar. From the data we're given we can use the approximations to calculate the period through the frequency and hopefully calculate the energy lost and magnetic field.

For Next Time

All in all, today we experimented with the libetempo and the functions it has to fit pulsar data and plot residuals and other data given .time and .par files. We also researched more characteristics we could find out about the pulsar and collected more information to put in the preliminary report.

For next time we need to:

- We need to start coding a bit out the data we have and see if we can create plots that are actually from the data set rather than from simulated data
- Work together to get on the same page about what we're investigating and plan/carry on writing the preliminary report
- Research more about the exact method we want to follow when analysing the data.

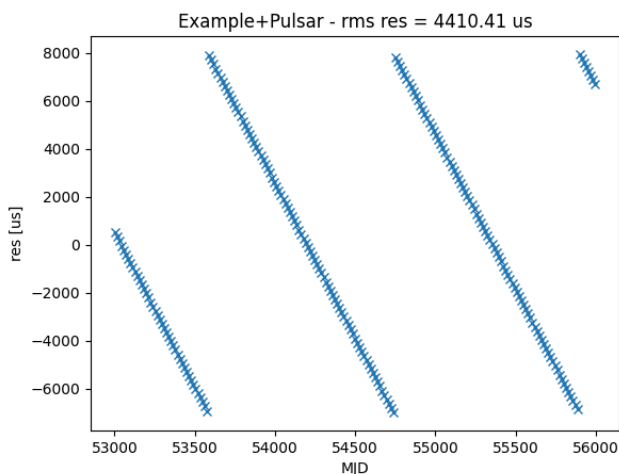
Monday 27th January 2025

To-Do Today:

- Get familiar with the functions provided by libstempo to analyse pulsar data, maybe attempt to use our real data to get 1 or 2 plots or values just rough estimates.
- Main focus is writing the preliminary report, finding more sources to support our research.

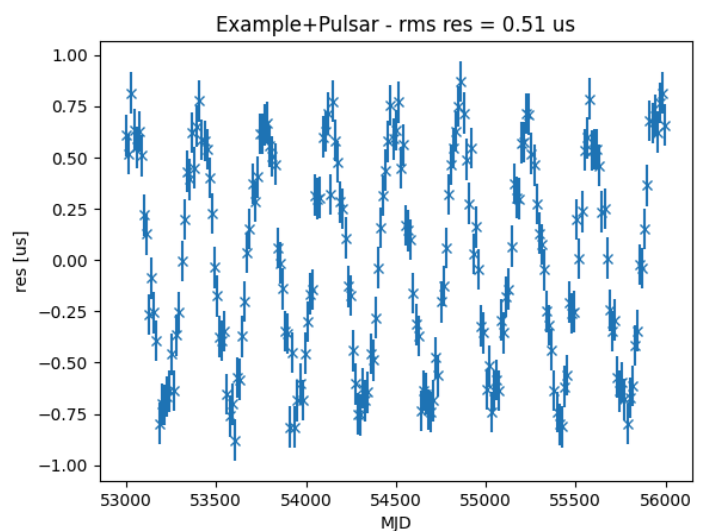
Work Achieved Today:

Today, I worked on getting familiar with the functions and precision of timing models. From the example pulsar, I managed to follow the tutorial and see how 1% error on some values creates very distinct plots. Shown below are some examples:

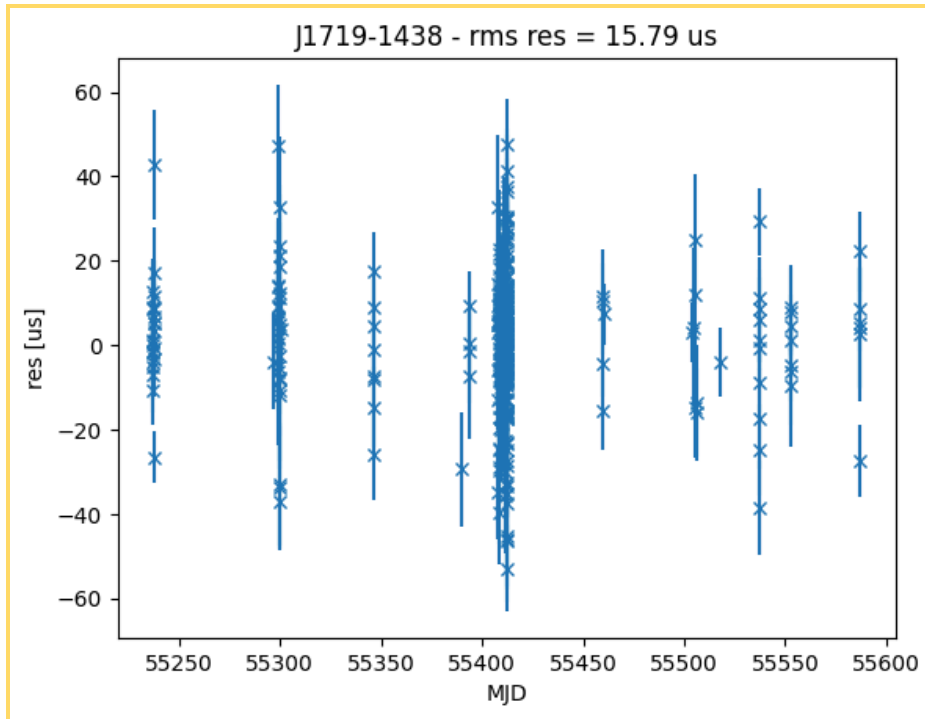


The first shows a $1e-7$ change in the frequency, a poor model showing inconsistent residuals to the test timings.

The second one below shows the 1% change in the declination. This model is a periodic wave that shows the effect of the Earth's orbit, causing a periodic variation every 364 days.



Then, once I was confident I managed to plot our data using the libstempo as shown in the figure below, showing a very spread data set. With small change in residuals of the order 10^{-6} .



From this we can see clear gaps which suggest breaks in observations where the pulsar wasn't observed, we can see that there are a lot of observations during the area between MJD 55410 -55415.

For Next Time:

After achieving some preliminary plots that we could show in our report, our plan for the next session is to mainly write/finish our preliminary report. If we finish early we could:

- Now that we had a preliminary plot of actual data, we wanted to see what could be done/shown from it, the very large band previously mentioned had to have some indication towards the period.

Thursday 30th January 2025

To - do Today:

- Today's main aim for most of us to finish our preliminary report that's due
- See if we can start working out a method towards finding that this plot was showing a companion orbiting it
- Perhaps work out some values about the pulsar that doesn't require the orbital period

Work Done Today:

Today we just focused on writing and reading our preliminary report. As a group, we were talking about how the timing model works, what preliminary plots suggest as to where we hope to move on from. This was all talked about to include in our report, but it was also a break to think about what lay ahead of us. We asked for help around how to cite websites and what to include in our report.

For Next Time:

Today was a light session in terms of progress, we just discussed what to include in our preliminary report such as equations, figures and papers that we could reference.

For next time we need to:

- Start working on refining our data as the data we have is very clumped together, so we need to work on it to show plots of orbital phase and oscillations.
- Maybe work out some of the characteristics of the pulsar that were included in the original paper as 'derived values' using values found in the .par file

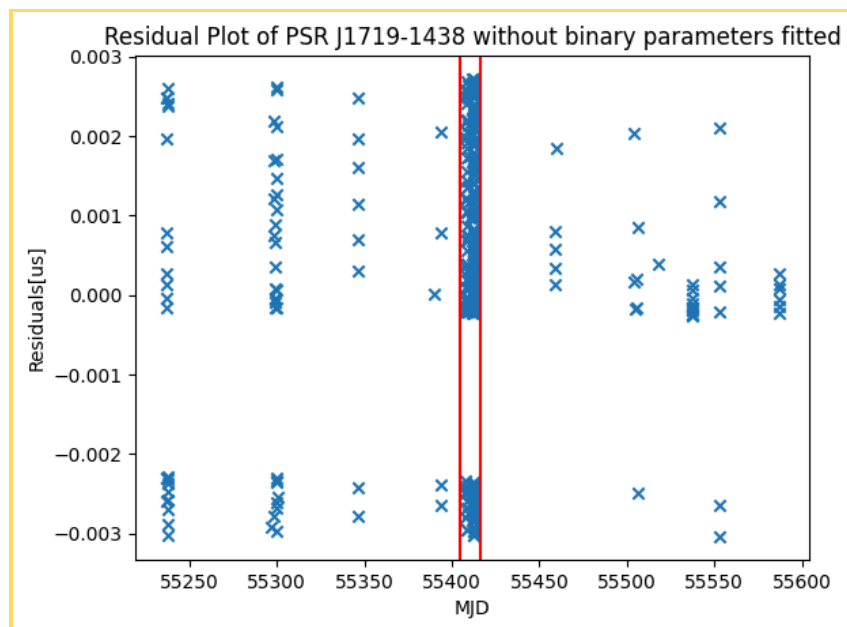
Monday 3rd February 2025

To - do Today:

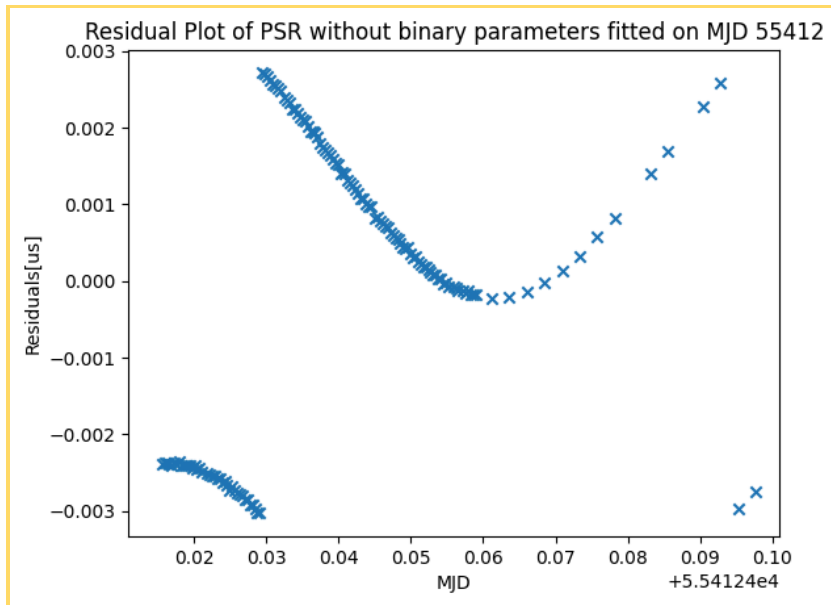
- Start to work on refining our data to show plots that actually help figuring out the period of the companion planet.
- Maybe start working on deriving some parameters about the pulsar, i.e surface magnetic field, or energy lost.

Work Done Today:

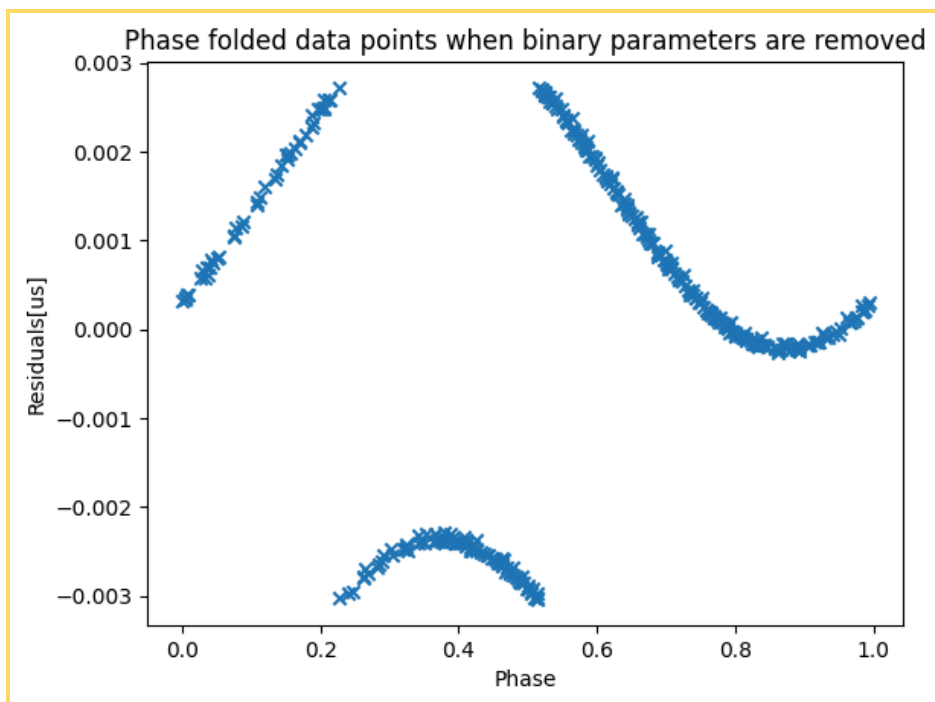
Today we made substantial progress towards finding the period of the companion planet. We began by using pyplots to better show the data, so we could narrow some more data points to a smaller time period. We also removed the binary section within the .par file as this meant that the fitted data we originally had would better show the data as the timing model built into Tempo2 won't correct the model for binary parameters. From this we can see our data more clearly, and we zoomed into the middle band which had more data points.



Zooming into the plot for this thick band, our plots now show a clear wave-like function with some errors. The residual on this graph 'wraps around' which effectively is trying to predict the pulse arrive but thinks the last pulse was late but was in fact early for the current one, so data starts at below -0.02 instead of above 0.03. So we need to work around and solve how to fix that.



We then used a fold function ([Appendix 8](#)) similar to how they use light curves which folds the data according to what position it is in the period. We did this for the whole data set and got a very nice graph showing one orbital phase for the planet.



This essentially proves the planet exists because all the data has been folded to its period, and provided an exact sinusoidal graph (peak is shifted downwards so need to fix that).

Next Time

For next time we need to:

- Describe a function that wraps the data around which essentially says that if the residual is below -0.02 it needs to add the period of the pulsar to it so that data can clearly show a sinusoidal wave
- Find a way to use the data to calculate the period of the planet, rather than using the period to show its there

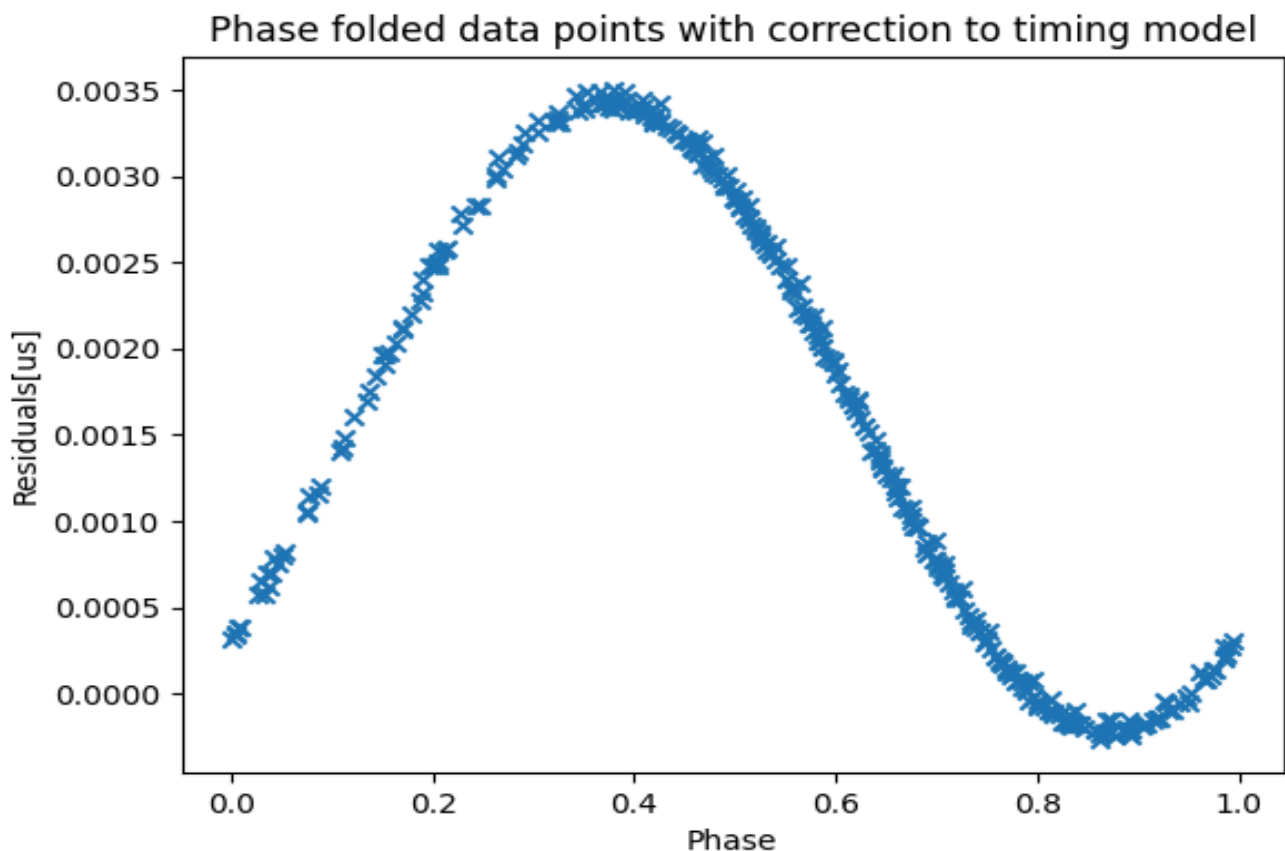
Thursday 6th February 2025**To-do Today:**

- Describe a function that wraps the data around which essentially says that if the residual is below -0.02 it needs to add the period of the pulsar to it so that data can clearly show a sinusoidal wave
- Now that the planet has been proven to exist using a phase fold that shows all the data is periodically repeating the pattern simulating an orbit. Need to find a way to get this period value

Work Done Today:

So to begin I began by defining a function to wrap around the data that was below -0.002 given by the function:

```
def wrap_around(residual, period): return np.where( residual < -0.002, residual + period, residual)
```

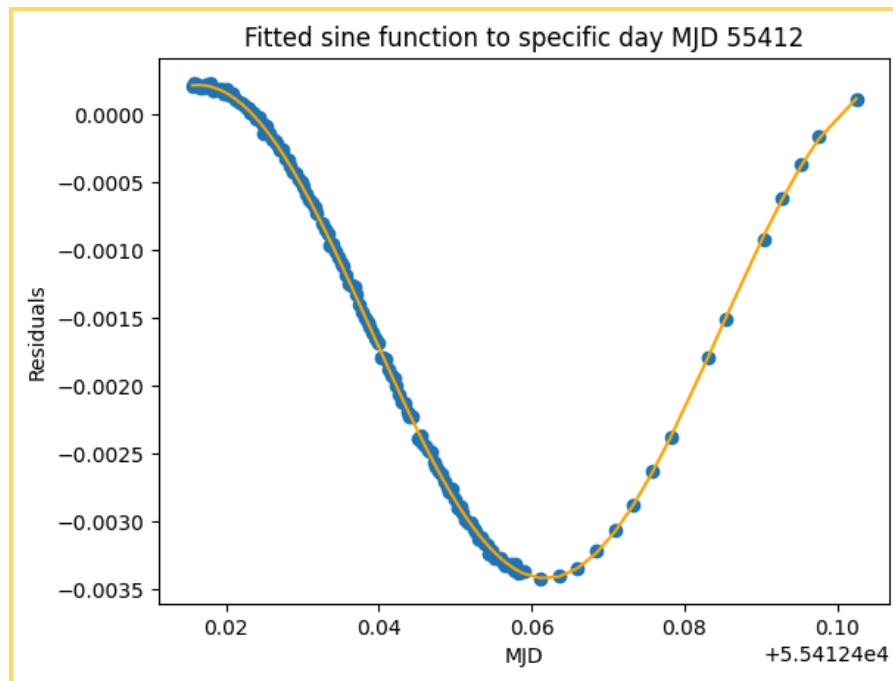


This warped our data nicely and now we can see a very evident sinusoidal wave for all the data points.

From the bottom graph we can see the full effect of the planet's 'tug' on the pulsar during one orbital phase. The only problem with this plot is that using the fold function requires us to know the period of the planet beforehand, so this plot is only useful after, but using a similar idea we can use an optimising function to find it for us.

So for our next step we defined a general sine wave, [Appendix 9](#), where A is the amplitude of the wave, b is the period of the wave, c is the horizontal offset and d is the vertical offset. We used this function for a limited data set. The full data set has too many breaks in time so we focused on MJD 55412 where we know the period of the planet could be seen. For this, we also had to change our wrap around function as the residual had flipped around and we need data >0.002 to shift downwards.

To find our values we used `scipy.optimize` which uses the fitting of least squares to find values. To get a better fit for the data we also included some base values that we assumed from the original graph such as period of 0.09 and amplitude of -0.001 by using the $(\text{argmax}-\text{argmin})/2$. This returned a fit as shown below:



with a period value of $P=0.09099982436980383\pm0.00014844934863069898\text{s}$ which is outside our value, we are roughly 2σ away.

For Next Time:

Today was successful in trying to find the period of the planet using optimising functions, but our value is too far from the actual value to consider as a guaranteed success before we can move on. So for next time:

- Either find a better method to optimise finding the period of the planet using the data or repeat method for other data and get an average to get a more accurate answer
- Start thinking about the presentation, delegation to everyone for their respective topic/ideas.

Monday 10th February 2025

To-Do Today:

- Try different methods to find a better estimate of the period of the planet
- Split into groups to be more efficient in finding more characteristics about the system i.e mass, magnetic field strength
- Start working on the presentation

Work Done today:

Today was mainly aimed towards trying to find a better method for estimating the period of the planet. It was a lot of brainstorming and trial and error. The idea of splitting the data into chunks and getting a weighted average didn't seem effective as a lot of the data was so incomplete that it couldn't even show half a wave properly. So we kept thinking of better ways to get a better fit. Until we ended up on least-squares fitting. Our aim is to use a sort of least squares fitting for the fold function. Since we know the value we are aiming for we keep fitting the fold function to the data until we get one full phase.

The idea compares multiple trial periods to each other and calculates the better one using chi-squared testing ([Appendix 10](#)). The lower the value, the better the fit, the model of chi-squared came from [Cochran 1952](#)

Meanwhile, using the initial approximation of the planet's period, some values were starting to be calculated using equations such as Kepler's law ([Appendix 11](#)) and mass function ([Appendix 12](#)).

For Next time:

Today's session was mainly brainstorming better ideas for getting a more accurate value of P, and now that we have a general idea to work towards, next session:

- To try and begin coding this least squares function for the phase fold function
- Continue working to provide plots for the presentation

Thursday 13th February 2025

To-Do Today:

Just work on the presentation

Work Done Today:

I was absent from the session today due to not feeling well so I just worked on the presentation whilst I was away, and made my current coding work more easier to read to produce plots with titles, labels and legends.

For Next time:

- Since I was absent I will catch up on the progress made on today's session, which was attempting the least squares fit function to get a more accurate value of P
- Finalise and finish up the presentation

Monday 17th February 2025**To-do Today:**

- Catch up on the work missed
- Complete full presentation
- Code any small data needed for the presentation

Work Done Today:

I caught up on the work, which included understanding how my group managed to code a least squares algorithm for the phase folding function so that we could calculate the period of the planet accurately. They managed to obtain a value of 0.090706 which is much more accurate to the true value than 0.0909. However this value still isn't accurate as we want it to be, so to improve this we planned to iterate over a smaller step length. A limitation was the length this would have to run for.

Our main objective today was to finish the presentation and allocate roles for each slide. So while making my slides. I also wanted to derive some more parameters about the pulsar that was given in Bailes' paper. The main one being the spin- down luminosity. I found this [website](#) which broke down the assumptions and equations needed to calculate the value along with other derivations.

From the assumption we could assume a mass of the pulsar of roughly 1.4 solar masses and we could obtain a radius using Kepler's laws. The spin down luminosity or otherwise known as the rate of energy loss due to release of radiation is given by the formula ([Appendix 3](#)), Where I is the moment of inertia of the pulsar. We can also assume that to be one of a uniform sphere so $I \sim 10^{45} \text{ g cm}^2$

From this I obtained a value of the spin down luminosity as $1.50\text{e}+33 \text{ erg/s}$ or as given in solar luminosities in the paper $0.390 \approx 0.40L_{\odot}$ which almost accurately matches the value in the paper.

For Next Time:

- Now that we have a better estimate for the period of the planet, we can go back to adjust some of our calculations about the planet's parameters.
- We also want to start to estimate some errors on our values using the error in the parameter files.
- For further analysis our last value we want to find is the radius of the planet and its density. We have read some papers on how to do this but still need to research more.

Monday 24th February 2025**To-Do Today:**

- One of today's aim is to review the least squares fit function and see how to improve it to better estimate the period as well as its error
- Error propagation and more derived calculations, for the values I calculated about the pulsar, I want to error propagate them so we can get relations to Bailes' paper

Work Done Today:

I started by making a collated colab document so everyone could access the code freely and find the values and figures for the final report when they wanted to. I spent some time cleaning up the code and explaining each bit in more detail.

Then I moved onto error propagation, I left Henry to complete the least squares and its errors as he was well into researching it, whilst I started on the error propagation for the derived pulsar values. Before I moved to that I wanted to make sure my values were accurate so I looked at this [website](#) about Pulsar as essential radio telescopes. This website broke down each characteristic and its derivation in detail with approximations.

The main takeaway was that for calculations we can assume our pulsar is a canonical pulsar of with $R=10 \text{ km}$, $M=1.4M_{\odot}$ and so $I \approx 10^{45} \text{ g cm}^2$. From this I calculated estimates for the upper radius of the pulsar ([Appendix 1](#)), rotational energy ([Appendix 2](#)), etc.

For values I got:

$$R < 54.03642506800112 \text{ km}$$

$$E = 6.556073041581962 \times 10^{43} \text{ J}$$

$$I = 1.1 \times 10^{38} \text{ kg m}^2 \text{ in less conventional units than gscm}$$

For Next Time

- I didn't get to calculate the actual errors, I read through derivations and understood the equations and theory needed for this. So still need to calculate errors
- As well as that continue doing more of the derivations to understand the pulsar's structure which could help understand why planets aren't common around them and the dynamics a planet might have with the pulsar

Thursday 27th February 2025

To-Do today:

- Error propagation and more derived calculations, for the values I calculated about the pulsar, I want to error propagate them so we can get relations to Bailes' paper
- As well as that continue doing more of the derivations to understand the pulsar's structure which could help understand why planets aren't common around them and the dynamics a planet might have with the pulsar

Work Done Today:

For today, my main focus is error propagation for every single value I've calculated. To do this I'm going to use the error propagation formula ([Appendix 13](#)).

I did this for most of the values, as some proved harder than others due to multiple variable differentiation. I managed to work through about half the calculated values. In the end I got errors for the period and period derivatives which weren't hard values to obtain. I also got values for the maximum

radius of the pulsar: 54.03642 ± 0.00057 km. As well as the the moment of inertia for the canonical pulsar as $I = 1.1135095275909085e+38 \pm 2.5025310390398435e+33$ kg m² which is essentially the $I \approx 1045$ g cm². I also got $E = 6.556073041581962e+43 \pm 1.4734293577413345e+39$ Joules, as the rotational energy of the pulsar, which led to the derivation of the spin down luminosity as 0.44 ± 0.03 $L_{\odot}$ which means that our value is just under 1σ away from Bailes paper value of $0.40(4)$ $L_{\odot}$.

For Next Time:

- Continue error propagation on more values such as the characteristic age and surface magnetism.
- Also attempt to derive a value for dispersion measure distance, and find its meaning as it was also stated in Bailes' paper so another comparison

Monday 3rd March 2025**To-Do Today:**

- Continue error propagation for the values I calculated about the pulsar, I want to error propagate them so we can get relations to Bailes' paper
- Also attempt to derive the Dispersion measure relation and its significance.

Work Done Today:

I continued on from last session's work of using the error propagation formula as shown below and calculating them for values I have so see its accuracy to true values.

I first began by doing this for the characteristic age ([Appendix 6](#)), which I previously mentioned in the presentation was incorrect possibly due to assumptions about the pulsar. However, it seems I forgot to convert the units appropriately from seconds to years. So to correct the formula I divided by a year, which gave me that the approximate age of the pulsar is $t > 12.5 \pm 0.775$ Gy which accurately matches the range in Bailes' paper.

Continuing on I obtained that the minimum surface magnetic field of the pulsar $B_s > 2.1e+08 \pm 6.5e+06$ G. This value is somewhat correct, the only discrepancy is that I calculated this as the minimum whereas in Bailes' paper it is the maximum. So in the coming sessions I need to investigate this.

Investigating typical ages and pulsars, found interesting theories about our system. The older the pulsar the slower it spins, however ours is very old and still is a millisecond pulsar. The magnetic field of our pulsar is also several orders smaller than a typical pulsar. This matched the requirements of a recycled pulsar. This theory suggests that when two stars are born in close proximity and one of them goes supernova they are either blown away or sometimes remain. If the latter occurs then as the second star gets older and starts to swell its outer layers, the pulsar/neutron star can start to absorb the outer layers leaving behind a dwarf star. This matches our system and is even mentioned by Bailes et al. , to prove this further calculations of the planet's parameters such as the density and radius could prove this.

I ended this session by deriving a new value that is also mentioned in Bailes' paper known as the Dispersion measure distance. I watched this [video](#) about radio astronomy and found a formula ([Appendix 7](#)). This is essentially finding the distance to the pulsar using the dispersion of radio waves through the interstellar medium. The DM value is found in the parameter file, having no error on it as $DM = 36.766 \text{ pc cm}^{-3}$, we then divide by the electron density of space, which in the video we took as approximately 0.03 cm^{-3} to ultimately get a value of $D = 1.2255333333333334 \text{ kpc}$ which is approx 1.23 kpc which is identical to the value in Bailes's paper.

For Next Time:

- I want to continue error propagation on more values. I have finished the derived values of the pulsar, now I will be helping Will with the parameters about the planet using the final estimation of the period.

Thursday 6th March 2025**To-Do today:**

- Help Will with error propagation on parameters on the planet using the accurate period value we have.

Work Done Today:

Using the most accurate value we could obtain of the period of the planet using the least squares fit, accomplished by Henry, we got that $P_b = 0.09070620869409898 \pm 2.963461571415812 \times 10^{-8}$ days which means our value is approximately 2.4σ away. To continue error propagation we began by calculating the radius of orbit, mass function, and mass of the planet and all their errors appropriately using ([Appendix 13](#)).

The orbital radius of the planet is $0.004419791040347073 \pm 4.683524259823267 \times 10^{-8} \text{ AU}$

The mass function is $1.5617542785440618 \times 10^{21} \pm 3.51029910930964 \times 10^{16} \text{ kg}$

The mass function is $7.854287496550158 \times 10^{-10} \pm 2.4964925278173328 \times 10^{-14} M_{\odot}$

The mass of the planet is $2.2959382258609322 \times 10^{27} \pm 3.846083062092732 \times 10^{22} \text{ Kg}$

The mass of the planet is $0.0011546604448583431 \pm 3.2365764123705275 \times 10^{-8} M_{\odot}$

The mass of the planet is $1.209582463176126 \pm 3.3905258351628923 \times 10^{-5} M_J$

Which we could compare to true values found on the internet such as the orbital radius on NASA's exoplanet archive as 0.0044 AU and well as a mass of 1.2 Mass of Jupiter, all of which accurately matches our calculated values.

For Next time

- Estimate the radius of the planet, this is a hard objective as the formula aren't certain and we have to using limits on the density which is obtained from relations about the orbital radius and the roche limit
 - Calculate statistical distances for our values to put in our final report.
-

Monday 10th March 2025

To-Do today:

- Look over whole project and realise and parts we have missed that we aimed to do
- Research over Roche-Lobe, density-period relation to get a radius on the planet

Work Done Today:

Unfortunately, I was away on this session due to a family emergency. The group notified me of any progress made with the penultimate session. The session was mainly used to start the report, collect references for the work we had done, and research towards the radius of the planet.

For Next time

- Finalise results making sure it is consistent with everyone, and uploaded on the collated work space so everyone can freely access data and plots.
 - See if the last session is enough to get a value for the radius/density of the planet
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Thursday 13th March 2025

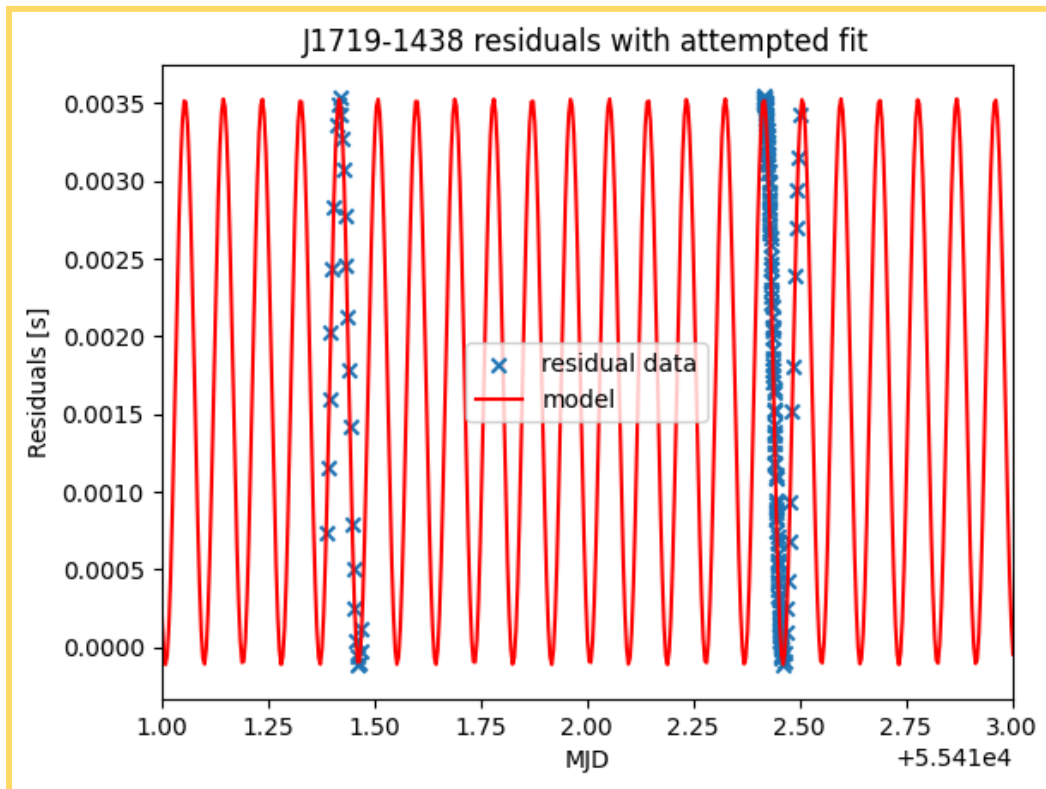
To-Do today:

- Finalise results making sure it is consistent with everyone, and uploaded on the collated work space so everyone can freely access data and plots.
- See if the last session is enough to get a value for the radius/density of the planet

Work Done Today:

After catching up with the group on the progress. I looked over the code and familiarised myself with the method of least squares and chi-squared testing that was used to calculate the period of the planet. Henry explained to me what the code was doing. The final result of this was a periodic wave with all the

optimal parameters. Creating a nice wave that went through all the data points.



However when I ran my code, the value obtained was different to his. The value obtained on the 6th was the value I was getting, however Henry obtained a closer value, so we sat there debugging the difference, as one was 0.77 standard deviations away the other was > 2 away. Eventually we narrowed it down to the parameter file, the difference rose from what parameters we deleted and counted as 'binary parameters'. So after realising I deleted some that didn't have to with the binary system, making the model more accurate. The final value orbital period was 0.09070627837842128 days.

In the end the calculations for the density and radius were not achieved, some approximates for the Roche lobe were made but not accurately enough. Some more research was definitely needed for this

There is no Next Time.

Appendix

$$R < \left(\frac{GMP^2}{4\pi^2} \right)^{1/3} \quad (1)$$

$$E = \frac{I\omega^2}{2} = \frac{4\pi^2 I}{P^2} \quad (2)$$

$$-\dot{E} = \frac{4\pi^2 I \dot{P}}{P^3} \quad (3)$$

$$P_{rad} = \frac{2}{3c^3} (BR^3 \sin \theta)^2 \left(\frac{2\pi}{P} \right)^4 \quad (4)$$

$$B > \left(\frac{3c^3 I}{8\pi^2 R^6} \right)^{1/2} (P\dot{P})^{1/2} \quad (5)$$

$$\tau = \frac{P}{2\dot{P}(365 \times 24 \times 3600)} \text{ years} \quad (6)$$

$$\text{Distance} = \frac{\text{DM}}{n_e} \text{ pc} \quad (7)$$

$$\phi = \frac{t - t_o}{P_b} \bmod 1 \quad (8)$$

$$x(t) = A \sin\left(\frac{2\pi}{P_b} t + C\right) + D \quad (9)$$

$$\chi_k^2 = \sum_{i=1}^N \frac{(t_i - x(t_i))^2}{\sigma_i^2} \quad (10)$$

$$a = \left(\frac{Gm_p P_b^2}{4\pi^2} \right)^{1/3} \quad (11)$$

$$f(m_c) = \frac{4\pi^2}{G} \frac{(a_p \sin i)^3}{P_b^2} = \frac{(m_c \sin i)^3}{(m_p + m_c)^2} \quad (12)$$

$$\sigma_F^2 = \left(\frac{\partial_F}{\partial_A} \sigma_A \right)^2 + \left(\frac{\partial_F}{\partial_B} \sigma_B \right)^2 + \left(\frac{\partial_F}{\partial_B} \sigma_B \right)^2 + \dots \quad (13)$$